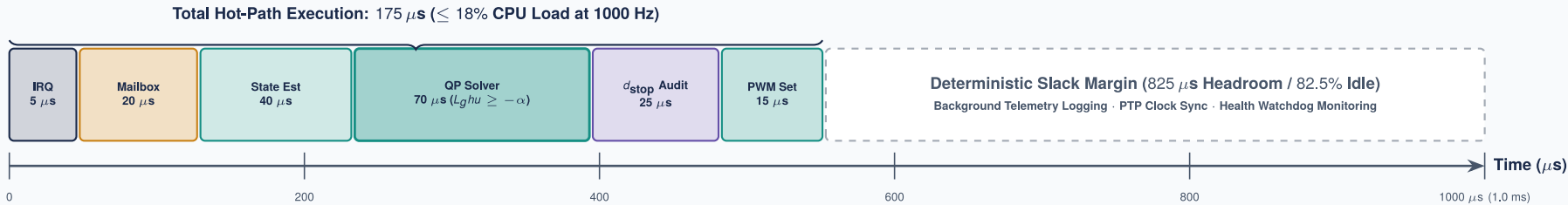


System 1 Real-Time Domain: Cortex-M4 Hard Real-Time 1.0 ms (1000 μs) ISR Slot

Guarantees: Zero Dynamic Heap Allocations (malloc = 0 B), TCM Static SRAM Buffers, Memory-Mapped Timer Registers.



The Linux MPU Reality (System 2): Non-Deterministic Scheduling Jitter

Why learned models running on application processors cannot hold direct electrical gate authority.



The Architectural Invariant:

- System 1 (Real-Time MCU): Hard 1000 Hz tick evaluates CBF invariants within < 200 μs, completely independent of Linux load.
- System 2 (Application MPU): Candidate action chunks stream asynchronously; late, stale, or jittered proposals are safely dropped.